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CSE 205

B.TECH. IVth SEMESTER EXAMINATION, 2025-26

BACHELOR OF TECHNOLOGY

CSE

Evolutionary Computing

Time : Three Hours] [Maximum Marks : 75

Note: There are **three** sections (A, B and C) and candidate has to attempt questions from all sections. Marks are indicated against each section.

Section-A

1. Answer **all** questions : 5×3=15

   (a) Differentiate between ANN and BNN with suitable diagram.

   (b) What is evolutionary Computing ? Explain the foundation of Evolutionary theory.

   (c) Explain various types of Mutation with suitable example.

(d) What is Hopfield Network ? Explain.

(e) Explain CSP and FOP with real life examples.

Section-B

Note: Answer **all** questions of the following : $4 \times 5 = 20$

2. (a) Explain the algorithm of PSO. Also explain the application of PSO.

Or

(b) What are evolutionary strategies ? Explain the importance of evolutionary computing algorithms in solving real life problems with suitable examples.

3. (a) Explain Supervised, Unsupervised and Reinforcement Learning with their suitable algorithms.

Or

(b) If the fitness value of four individual A, B, C, D are given as :

S.N	Individual	Fitness
1.	A	2
2.	B	4
3.	C	6
4.	D	8

Find the probability of Best fittest according to the Linear Rank based selection.

4\. (a) What is selection operator in GA ? Explain various types of selection operators.

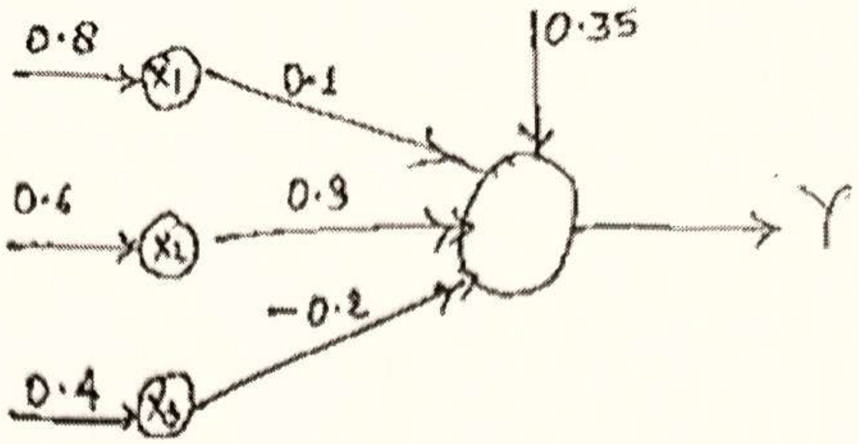
Or

(b) Find the output of the neuron Y using :

$\frac{1}{1-e^{x}}$

(i) Sigmoid activation function

(ii) Tanh activation function



5. (a) What do you understand by Multi-objective optimization in evolutionary computing Explain.

Or

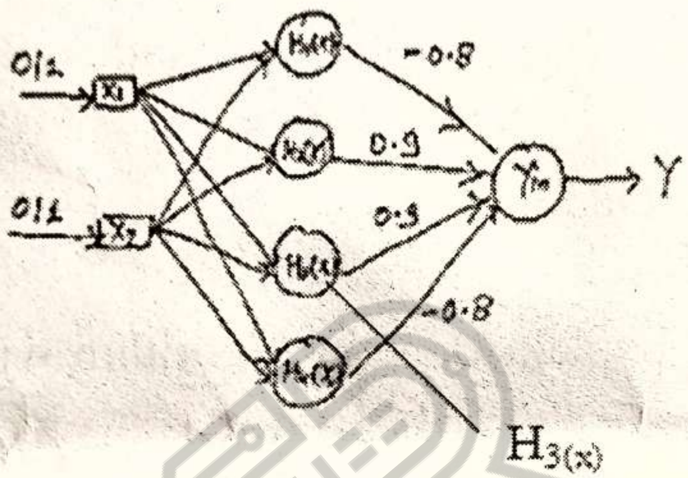
(b) What is activation function ? Explain various types of activation function in detail.

Section-C

Note: Answer any **two** questions of the following : 2×20=40

6. (a) Explain Honey-Bee Optimization with the help of algorithm and flowchart.

(b) Consider the XOR Boolean function that has 4 patterns (0,0) (0,1) (1,0) and (1,1) in a 2-dimensional input space. Construct a RBFNN as shown below for XOR Boolean function. Show each steps clearly.



7. (a) What is swarm intelligence ? Explain Ant colony optimization (ACO) with algorithm.

(b) Design a perception that performs the Boolean function AND and update the weight until the Boolean function gives the desired output.

Given - $W_1 = 0.3$, $W_2 = -0.2$

$\propto = 0.2$, bias $\theta = -0.4$

Activation function : Binary step

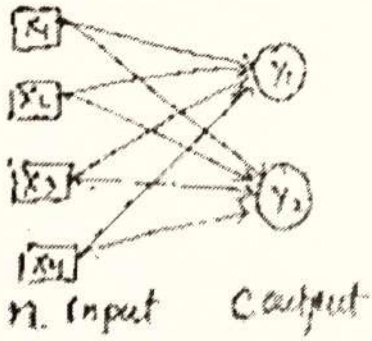
8. (a) If population size = 4 and initial population = {13, 14, 8, 19}, Perform genetic algorithm to maximize the function $f(x) = x^2 + 2$ over {0 - 31} with single point crossover, bitwise mutation and roulette wheel selection.

(b) Explain Genetic Algorithm with the help of flow chart. Also explain its application.

9. (a) Explain the role of a evolutionary computing in the field of Artificial intelligence with examples.

(b) Train the SOFM network by determining the class memberships of the input data. Show each steps clearly.

Training Samples :



$X_1 : (1, 0, 1, 0)$

$X_2 : (1, 0, 0, 0)$

$X_3 : (1, 1, 1, 1)$

$X_4 : (0, 1, 1, 0)$

Output Units : Unit 1 Unit 2

initial weight matrix

$$\begin{bmatrix} Unit\ 1 \\ Unit\ 2 \end{bmatrix} : \begin{bmatrix} 0.3 & 0.5 & 0.7 & 0.2 \\ 0.6 & 0.5 & 0.4 & 0.2 \end{bmatrix}$$

Learning weight θ = 0.6

1 0 0 1
8 4 2 1

2^2 2^1 2^0